



Gosford High School

2010
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time - 5 minutes.
- Working Time - 2 hours.
- Write using a blue or black pen.
- Board Approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question in a new booklet

Total marks (84)

- Attempt Questions 1-7.
- All questions are of equal value.

Question 1.

(Begin a new booklet)

a) Find the value of $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$ (1)

b) Find the acute angle between the lines $y = 3x - 2$ and $y = 2 - x$, giving your answer to the nearest degree. (2)

c) Solve $\frac{(x+1)}{x} > 0$. (2)

d) Find the number of ways in which 3 boys and 3 girls can be arranged in a straight line so that the tallest boy and tallest girl occupy the two middle positions. (2)

e) Find the value of k such that $(x - 2)$ is a factor of $P(x) = x^3 + 2x + k$ (2)

f) Evaluate $\int_0^{\pi} \sin^2 x \, dx$. (3)

Question 2.

(Begin a new booklet)

- a) The interval AB is divided externally in the ratio 1:4. If A and B are the points (1,3) and (6,-2) respectively, find the coordinates of the point of division. (2)

- b) Given that a root of $y = x + \ln x - 2$ lies close to $x = 1.5$, use Newton's method once to find an improved value of that root. (3)

- c) Using the substitution $u = x^2 - 2$ or otherwise, find $\int \frac{x}{\sqrt{x^2 - 2}} dx$. (2)

- d) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The equation of the tangents at P and Q respectively are $y = px - ap^2$ and $y = qx - aq^2$.

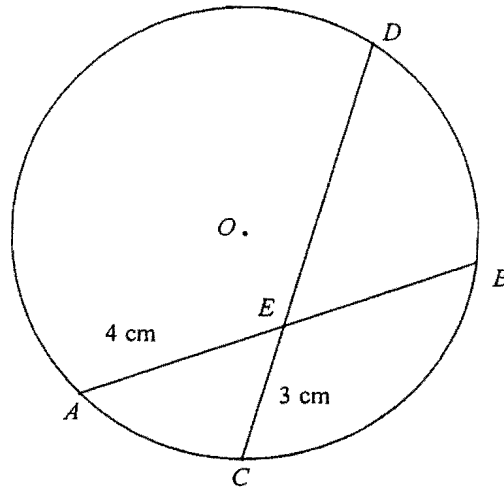
- (i) The tangents at P and Q meet at the point R. Show that the coordinates of R are $(a(p + q), apq)$. (2)

- (ii) The equation of the chord PQ is $y = \frac{p+q}{2}x - apq$
(Do NOT show this.) If the chord PQ passes through $(0, a)$, show that $pq = -1$. (1)

- (iii) Find the equation of the locus of R if the chord PQ passes through $(0, a)$ (2)

Question 3.**(Begin a new booklet)**

- a) In the circle centred at O , the chords AB and CD intersect at E .
The length of AB is x cm and of CD is y cm.
 $AE = 4$ cm and $CE = 3$ cm.



Show that $4x = 3y + 7$ (2)

- b) Consider the function $f(x) = \frac{3x}{x^2 - 1}$
- Show that the function is odd. (1)
 - Show that the function is decreasing for all values of x . (1)
 - Sketch the graph of the function showing clearly the equations of any asymptotes (2)
- c) If $\tan A$ and $\tan B$ are the roots of the quadratic equation $3x^2 - 5x - 1 = 0$, find the value of $\tan(A + B)$. (2)
- d) Solve $\cos 2A = \cos A$ where $0 \leq A \leq 2\pi$ (2)
- e) Write down the general solutions of $\tan(x - \frac{\pi}{3}) = -1$ (2)

Question 4.

(Begin a new booklet)

- a) At time t hours after an oil spill occurs, a circular oil slick has a radius r km, where $r = \sqrt{t+1} - 1$. Find the rate at which the area of the slick is increasing when its radius is 1 km, giving your answer correct to 2 decimal places. (3)
- b) Use Mathematical Induction to show that $3^n - 2n - 1$ is divisible by 4 for all positive integers $n \geq 2$ (4)
- c) At time t years after observation begins, the number N of birds in a colony is given by $N = 100 + 400e^{-0.1t}$
- i. Sketch the graph of N as a function of t showing clearly the initial population size and the limiting population size. (2)
- ii. Find the time taken for the population size to fall to half its initial value, giving the answer correct to the nearest year (2)
- d) Given $f(x) = \log_e(\sqrt{9-x^2})$, state the domain of $f(x)$. (1)

Question 5.**(Begin a new booklet)**

- a) Use the substitution $t = \tan \frac{x}{2}$ to show that $\operatorname{cosec} x + \cot x = \cot \frac{x}{2}$ (2)

Hence evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x) dx$ (3)

- b) Find the term independent of x in the expansion of $\left(x^2 - \frac{2}{x}\right)^9$ (3)

- c) Show, using sketches on separate sets of axes :

- (i) the area enclosed between $y = \sin^{-1} x$, the x axis, and the line $x = 1$ (1)

- (ii) the area enclosed between $y = \sin x$, the x axis, and the line $x = \frac{\pi}{2}$. (1)

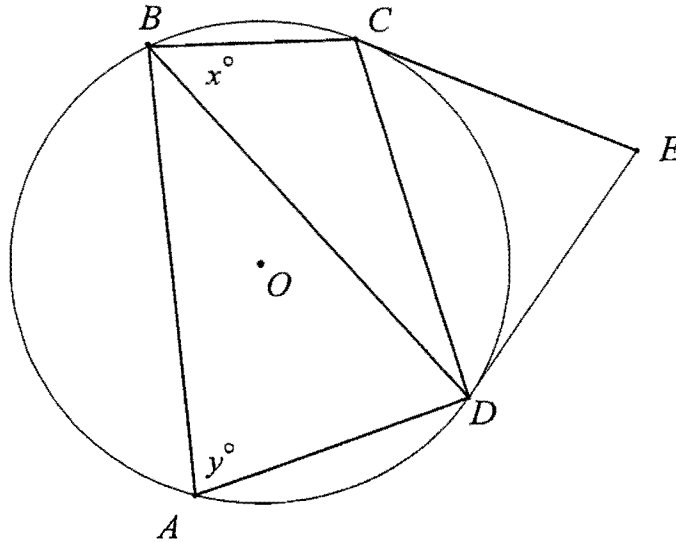
- (ii) Using the graphs, explain why

$$\int_0^1 \sin^{-1} x \, dx = \frac{\pi}{2} - \int_0^{\frac{\pi}{2}} \sin x \, dx. \quad (2)$$

Question 6.

(Begin a new booklet)

- a) The circle $ABCD$ has centre O . Tangents are drawn from an external point E to contact the circle at C and D . $\angle CBD = x^\circ$ and $\angle BAD = y^\circ$.



- i. Copy the diagram into your examination booklet
 - ii. Show that $\angle CED = (180 - 2x)^\circ$. (2)
 - iii. Show that $\angle BDC = (y - x)^\circ$. (2)
- b) A particle moving in a straight line is performing Simple Harmonic Motion. At the time t seconds it has displacement x metres from a fixed point O on the line, where $x = 4 \cos^2 t - 1$
- i. Show that its acceleration is given by $\ddot{x} = -4(x - 1)$ (2)
 - ii. Sketch the graph of x as a function of t for $0 \leq t \leq \pi$, clearly showing the times when the particle passes through O . (2)
 - iii. For $0 \leq t \leq \pi$, find the time when the velocity of the particle is increasing most rapidly, and find this rate of increase in the velocity. (2)
- c) Differentiate $2x^2 \cos^{-1} 2x$. (2)

Question 7.

(Begin a new booklet)

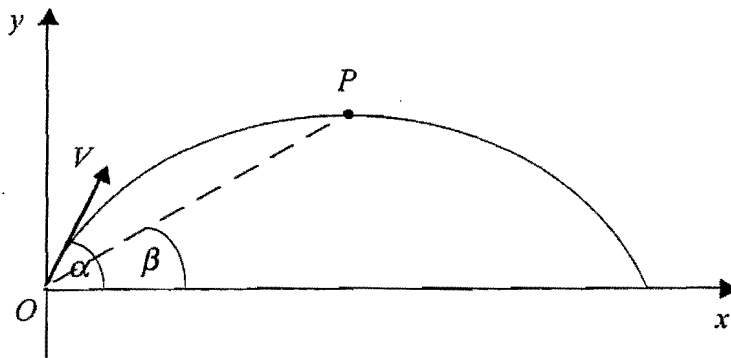
- a) The polynomial $P(x) = 2x^3 - 5x^2 - 3x + 1$ has zeros α , β and γ .
Find the values of

(i) $3\alpha + 3\beta + 3\gamma - 4\alpha\beta\gamma$ (2)

(ii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ (1)

(iii) $\alpha^2 + \beta^2 + \gamma^2$ (1)

b)



A particle is projected from a point O with speed $V \text{ ms}^{-1}$ at an angle α above the horizontal, where $0 < \alpha < \frac{\pi}{2}$. It moves in a vertical plane subject to gravity where the acceleration due to gravity is 10 ms^{-2} . At time t seconds it has horizontal and vertical displacements x metres and y metres respectively from O . At point P where it attains its greatest height the angle of elevation of the particle from O is β radians.

(i) Use integration to show that $x = Vt \cos \alpha$ and $y = Vt \sin \alpha - 5t^2$. (2)

(ii) Show that $\tan \beta = \frac{1}{2} \tan \alpha$. (3)

(iii) If the particle has greatest height 80 m above O at a horizontal distance 120 m from O , find the exact values of α and V . (3)

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$$\text{Q1/ (a) } \lim_{x \rightarrow 0} \frac{3}{2} \frac{\sin 3x}{3x} = \frac{3}{2} \times 1$$

$$= \frac{3}{2}$$

$$(b) \quad m_1 = 3 \quad m_2 = -1 \quad \tan \theta = \left| \frac{3 - (-1)}{1 + (3)(-1)} \right|$$

$$\tan \theta = \left| \frac{4}{-2} \right|$$

$$\theta = \tan^{-1}(2)$$

$$\theta \approx 63^\circ$$

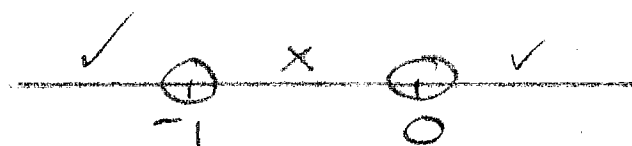
$$(c) \quad \frac{(x+1)}{x} > 0 \quad \text{CP at } x=0, x \neq 0.$$

Make an equation to find other CP's

$$\frac{x+1}{x} = 0$$

$$x+1=0$$

$$x = -1 \quad \text{CP at } x = -1, x \neq -1$$



$$x > 0, x < -1$$

$$(d) \quad \boxed{4} \boxed{3} \underbrace{\boxed{2} \boxed{2}}_{2!} \boxed{2} \boxed{1}$$

$$\therefore 2! \times 4! = 48 \text{ ways}$$

$$(e) \quad P(x) = x^2 + 2x + k$$

$$\text{But } P(2) = 0$$

$$0 = 2^2 + 2(2) + k$$

$$0 = 12 + k$$

$$k = -12$$

$$(f) \quad \int_0^\pi \frac{1}{2} (1 - \cos 2x) dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^\pi$$

$$= \frac{1}{2} \left[\pi - \frac{1}{2} \sin 2\pi - \left(0 - \frac{1}{2} \sin 0 \right) \right]$$

$$= \frac{1}{2} [\pi - 0 - (0 - 0)]$$

$$= \frac{\pi}{2}$$

Q2/(a) A B
 (1,3) $\rightarrow$ (6,-2)
 1 : -4

$$= \left(\frac{(1)(6) + (-4)(1)}{-3}, \frac{(1)(-2) + (-4)(3)}{-3} \right)$$

$$= \left(\frac{2}{-3}, \frac{14}{3} \right)$$

(b) $y = x + \ln x - 2$
 $y' = 1 + \frac{1}{x}$

$$f(1.5) = 1.5 + \ln 1.5 - 2$$

$$= \ln 1.5 - 0.5$$

$$f'(1.5) = 1 + \frac{2}{3}$$

$$= \frac{5}{3}$$

$$x_2 = 1.5 - \left[\frac{\ln(1.5) - 0.5}{\frac{5}{3}} \right]$$

$$= 1.56 \text{ (to 2 dec places)}$$

(c) $\int \frac{x}{\sqrt{x^2-2}} dx$

$$u = x^2 - 2$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$\frac{du}{2x} = dx$$

$$\therefore \int \frac{x}{\sqrt{u}} \cdot \frac{du}{2x}$$

$$= \frac{1}{2} \int u^{-\frac{1}{2}} du$$

$$= \frac{1}{2} \cdot 2 u^{\frac{1}{2}} + C$$

$$= \sqrt{u} + C$$

$$= \sqrt{x^2-2} + C$$

$$(d) (i) \quad px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p-q) = a(p-q)(p+q)$$

$$x = a(p+q)$$

$$y = px - ap^2$$

$$y = ap(p+q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

$$(ii) \quad y = \frac{(p+q)x}{2} - apq$$

$$\text{Sub } (0, a)$$

$$a = 0 - apq$$

$$a = -apq$$

$$-1 = pq$$

$$(iii) \quad x = a(p+q) \quad y = apq$$

$$y = -a$$

$$\therefore y = -a \text{ (is the locus)}$$

Q3/ $AE \cdot EB = CE \cdot ED$
 $4 \cdot (x-4) = 3 \cdot (y-3)$
 $4x - 16 = 3y - 9$
 $4x = 3y + 7$

(b) $f(x) = \frac{3x}{x^2-1}$

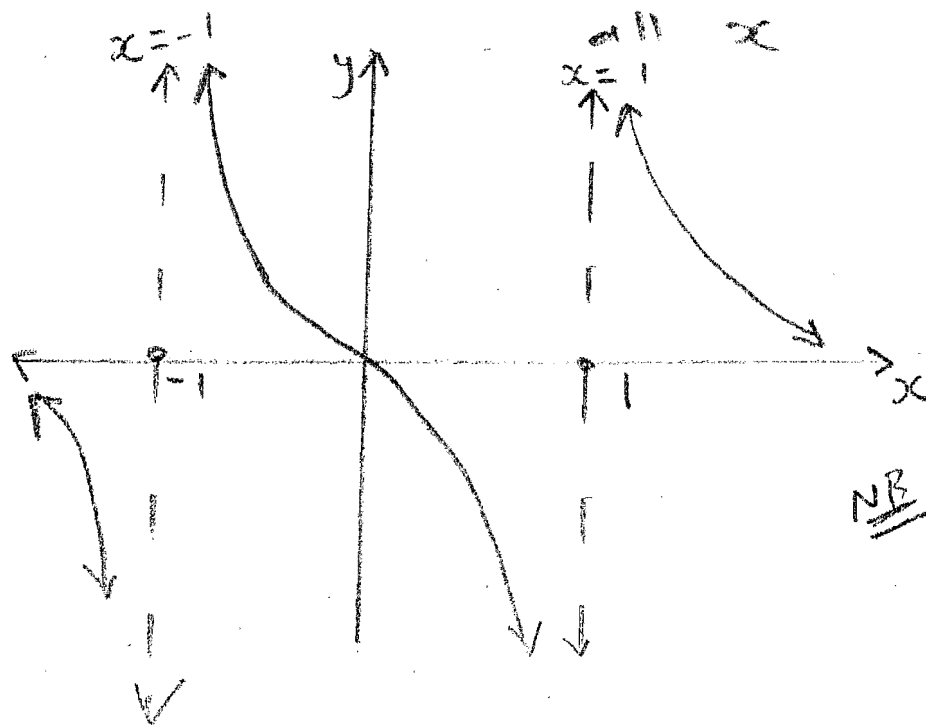
(i) $f(-x) = \frac{-3x}{x^2-1}$

$\therefore f(x) = -f(-x) \therefore$ odd

(ii) $f'(x) = \frac{(x^2-1) \cdot 3 - (x^2) \cdot 6x}{(x^2-1)^2}$
 $= \frac{3x^2 - 3 - 6x^2}{(x^2-1)^2}$
 $= \frac{-3x^2 - 3}{(x^2-1)^2}$

Now $(x^2-1)^2 > 0$ for all x
 $-3(x^2+1) < 0$ for all x

$\therefore \frac{+}{-} = - \quad f'(x) < 0$ for all x
 $\therefore$ function decreases for



NR no
st pts

$$(c) \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \quad 3x^2 - 5x - 1 = 0$$

$$\alpha + B = \tan A + \tan B$$

$$\frac{5}{3} = \tan A + \tan B$$

$$\tan A \tan B = \alpha B \\ = -\frac{1}{3}$$

$$\therefore \tan(A+B) = \frac{\frac{5}{3}}{1 - (-\frac{1}{3})} \\ = \frac{\frac{5}{3} \times \frac{3}{4}}{1} \\ = \frac{5}{4}$$

$$(d) \cos 2A = \cos A \quad 0 \leq A \leq 2\pi$$

$$2\cos^2 A - 1 = \cos A$$

$$2\cos^2 A - \cos A - 1 = 0$$

$$\begin{array}{cc} 2\cos A & 1 \\ \cos A & -1 \end{array} \quad \begin{array}{c} \diagup \\ \diagdown \end{array}$$

$$(2\cos A + 1)(\cos A - 1) = 0$$

$$2\cos A = -1 \quad \text{or} \quad \cos A = 1 \\ \cos A = -\frac{1}{2}$$

$$A = \frac{2\pi}{3}, \frac{4\pi}{3}, 0, 2\pi$$

$$(e) \tan\left(x - \frac{\pi}{3}\right) = -1$$

$$x - \frac{\pi}{3} = n\pi - \frac{\pi}{4}$$

$$x = n\pi - \frac{\pi}{4} + \frac{\pi}{3}$$

$$x = n\pi + \frac{\pi}{12}$$

Question 4

$$(a) \quad r = (t+1)^{\frac{1}{2}} - 1$$

$$\frac{dA}{dt} = ?$$

$$\frac{dA}{dt} = \frac{dr}{dt} \times \frac{dA}{dr}$$

$$A = \pi r^2$$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dr}{dt} = \frac{1}{2}(t+1)^{-\frac{1}{2}}$$

$$= 2\pi [(t+1)^{\frac{1}{2}} - 1]$$

$$= \frac{1}{2(t+1)^{\frac{1}{2}}}$$

$$\frac{dA}{dt} = \frac{1}{2(t+1)^{\frac{1}{2}}} \times 2\pi [(t+1)^{\frac{1}{2}} - 1]$$

but when $r=1$

$$1 = (t+1)^{\frac{1}{2}} - 1$$

$$2 = (t+1)^{\frac{1}{2}}$$

$$4 = t+1$$

$$t=3$$

$$= \frac{\pi [4^{\frac{1}{2}} - 1]}{4^{\frac{1}{2}}}$$

$$= \frac{\pi}{2} \text{ km}^2/\text{hr}$$

$$\approx 1.57 \text{ km}^2/\text{hr}$$

(b) Step 1 Prove true for $n=2$

$$3^2 - 2(2) - 1 = 9 - 4 - 1 \\ = 4$$

$\therefore$ true for $n=2$

Step 2 Assume true for $n=k$ where k is an integer ≥ 2

ie $3^k - 2k - 1 = 4M$ (where M is a positive integer)
 $3^k = 4M + 2k + 1$

Step 3 Prove true for $n=k+1$

ie $3^{k+1} - 2(k+1) - 1 = 3^{k+1} - 2k - 2 - 1 \\ = 3^{k+1} - 2k - 3$
(is divisible by 4)

Now $3^k \cdot 3 - 2k - 3 = 3 \cdot (4M + 2k + 1) - 2k - 3$
FROM ASSUMPTION
 $= 12M + 6k + 3 - 2k - 3 \\ = 12M + 4k \\ = 4(3M + k)$
which is divisible by 4

Step 4 As it is true for $n=2$ and if true for $n=k$, it is true for $n=k+1$ and all positive integers of $n \geq 2$.

$$\begin{aligned}
 \text{(iii)} \quad x^2 + y^2 + z^2 &= (x+y+z)^2 - 2(xy+yz+xz) \\
 &= \left(\frac{5}{2}\right)^2 - 2\left(-\frac{3}{2}\right) \\
 &= \frac{25}{4} + \frac{12}{4} \\
 &= \frac{37}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \ddot{x} &= 0 \\
 \dot{x} &= \int 0 \, dt \\
 &= C \quad \text{when } t=0 \quad \dot{x} = v \cos \alpha
 \end{aligned}$$

$$\begin{aligned}
 \therefore \dot{x} &= v \cos \alpha \\
 x &= \int v \cos \alpha \, dt
 \end{aligned}$$

$$\begin{aligned}
 x &= vt \cos \alpha + C \quad \text{when } t=0 \quad x=0 \\
 \therefore C &= 0
 \end{aligned}$$

$$x = vt \cos \alpha$$

$$\begin{aligned}
 \ddot{y} &= -10 \\
 \dot{y} &= \int -10 \, dt
 \end{aligned}$$

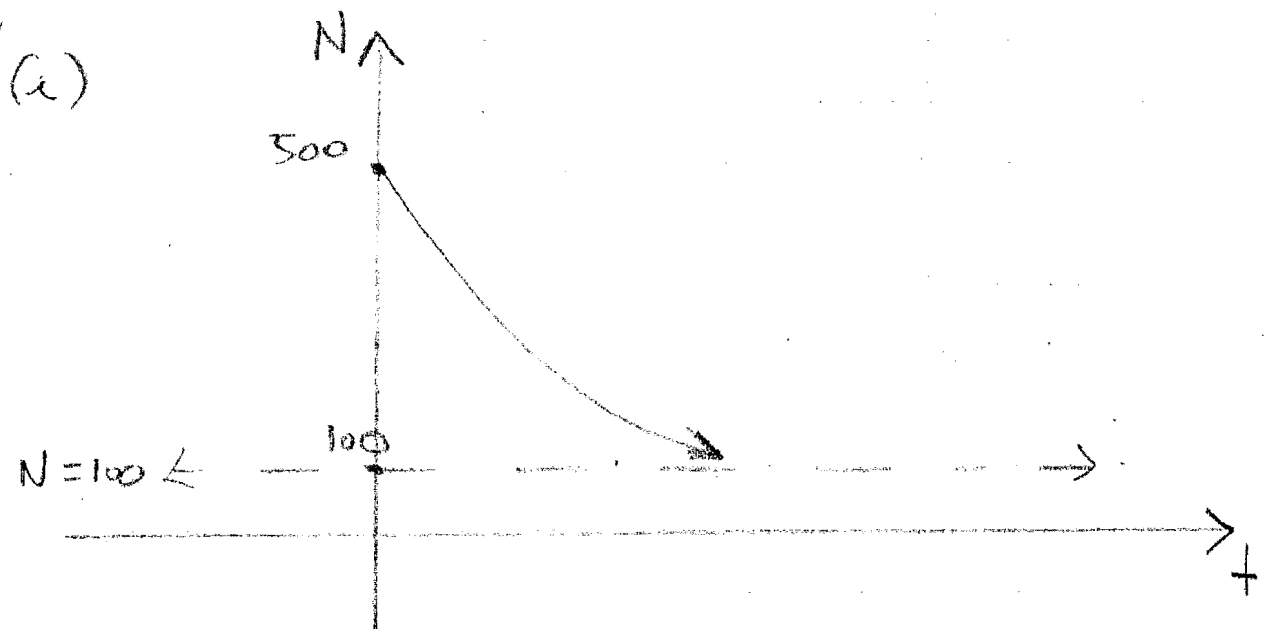
$$= -10t + C \quad \text{when } t=0 \quad \dot{y} = v \sin \alpha$$

$$\begin{aligned}
 \therefore \dot{y} &= -10t + v \sin \alpha \\
 y &= \int -10t + v \sin \alpha \, dt
 \end{aligned}$$

$$= -5t^2 + vt \sin \alpha + C \quad \text{when } t=0 \quad y=0$$

$$\therefore y = vt \sin \alpha$$

$$(c) \quad N = 100 + 400 e^{-0.1t}$$



$$(ii) \quad 250 = 100 + 400 e^{-0.1t}$$

$$\frac{150}{400} = e^{-0.1t}$$

$$\ln\left(\frac{3}{8}\right) = -0.1t$$

$$t = \left(\frac{\ln 3 - \ln 8}{-0.1} \right)$$

$$t = 9.808 \text{ years}$$

$\therefore$ After 10 years

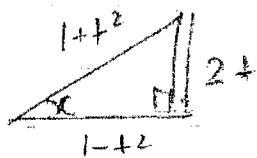
$$(d) \quad f(x) = \log_e \sqrt{9-x^2}$$

$$\therefore \sqrt{9-x^2} > 0$$

$$\therefore \text{Domain is } -3 < x < 3$$

Question 5

(a)



$$t = \tan \frac{x}{2}$$

$$\cot \frac{x}{2} = \frac{1}{t}$$

$$\operatorname{cosec} x = \frac{1+t^2}{2t}$$

$$\cot x = \frac{1-t^2}{2t}$$

$$\operatorname{cosec} x + \cot x = \cot \frac{x}{2}$$

$$\text{LHS} = \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$$

$$= \frac{2}{2t}$$

$$= \frac{1}{t}$$

$$= \cot \frac{x}{2}$$

$$\therefore \text{LHS} = \text{RHS}$$

(b)

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot \frac{x}{2} dx = 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\frac{1}{2} \cos \frac{x}{2}}{\sin \frac{x}{2}} dx$$

$$= 2 \left[\ln \left(\sin \frac{x}{2} \right) \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= 2 \left[\ln \left(\sin \frac{\pi}{4} \right) - \ln \left(\sin \frac{\pi}{6} \right) \right]$$

$$= 2 \left[\ln \frac{1}{\sqrt{2}} - \ln \frac{1}{2} \right]$$

$$= 2 \ln \left[\frac{1}{\sqrt{2}} \times \frac{2}{1} \right]$$

$$= 2 \ln \sqrt{2}$$

$$= 2 \times \frac{1}{2} \ln 2$$

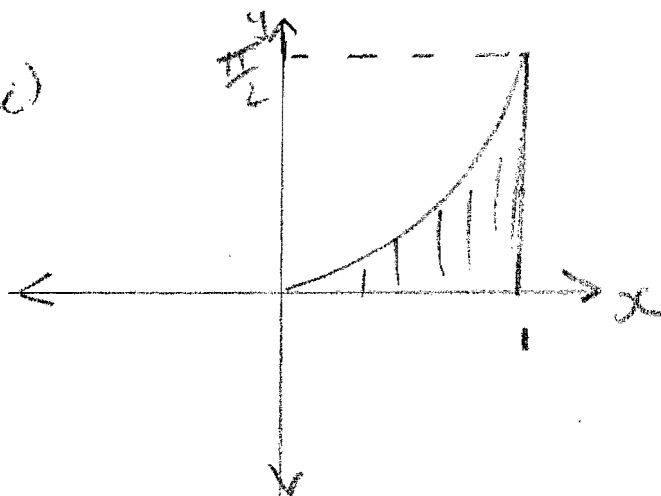
$$= \ln 2$$

$$\begin{aligned}
 (b) \left(x^2 - \frac{2}{x}\right)^9 &= {}^9C_K (x^2)^{9-K} \left(-\frac{2}{x}\right)^K \\
 &= {}^9C_K x^{18-2K} \cdot (-2)^K x^{-K} \\
 &= {}^9C_K (-2)^K x^{18-3K}
 \end{aligned}$$

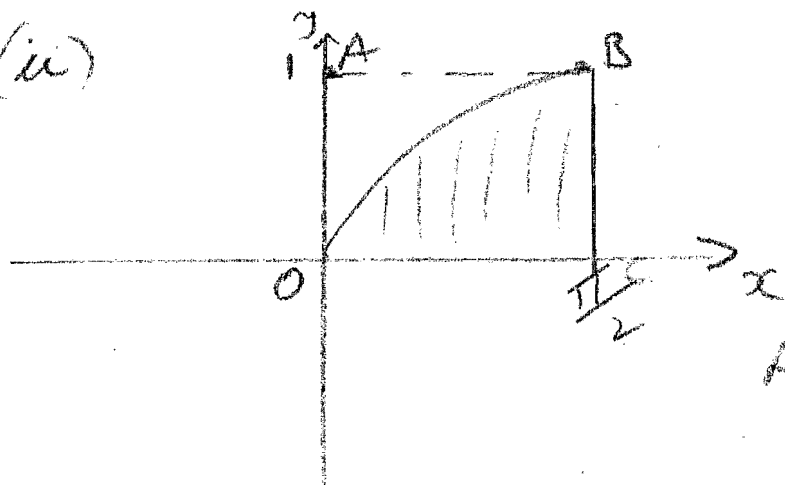
Independent of x when $18-3K=0$
 $18=3K$
 $K=6$

$$\therefore {}^9C_6 (-2)^6 = 5376$$

(c) (i)



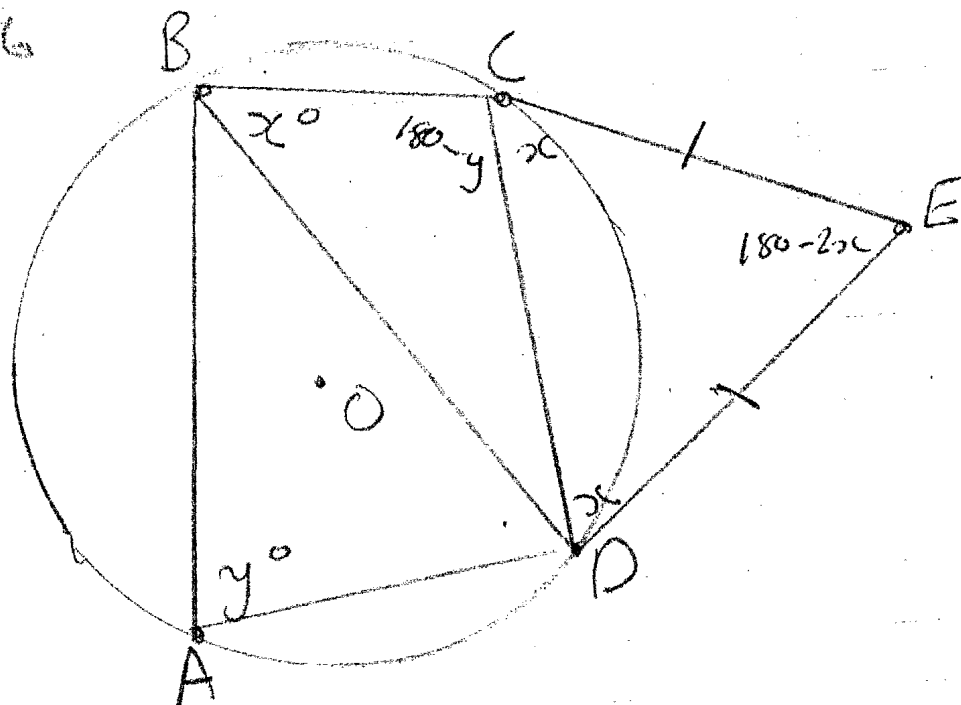
(ii)



(iii) $\int_0^1 \sin^{-1} x \, dx$
 corresponds to
 the area OAB
 shown in part (ii)
 due to symmetry.

Thus
 Area of rectangle OACB
 minus $\int_0^{\pi/2} \sin x \, dx$
 gives $\int_0^1 \sin^{-1} x \, dx$

Question 6
(a)



(i) $\angle EDC = \angle DBC = x^\circ$ (alternate segment theorem)

$CE = DE$ (equal tangents from point E)
 $\angle EDC = \angle DCE = x$ (base $\angle$'s isosceles $\triangle CED$)

$\therefore \angle CED = 180 - 2x$ ($\angle$ sum $\triangle CED$)

(ii) $\angle BCD = 180 - y$ (opp $\angle$'s cyclic quad ABCD are supplementary)

$\angle BDC = 180 - (180 - y) - x$
 $= y - x$ ($\angle$ sum $\triangle BDC$)

$$(b) \quad x = 4 \cos^2 t - 1$$

$$(i) \quad \dot{x} = 4 \cdot 2(\cos t) \cdot -\sin t \\ = -4 \sin 2t$$

$$\ddot{x} = -4 \cdot 2 \cos 2t$$

$$\text{but } x = 4 \cos^2 t - 1$$

$$\cos 2t = 2 \cos^2 t - 1$$

$$2 \cos 2t = 4 \cos^2 t - 2$$

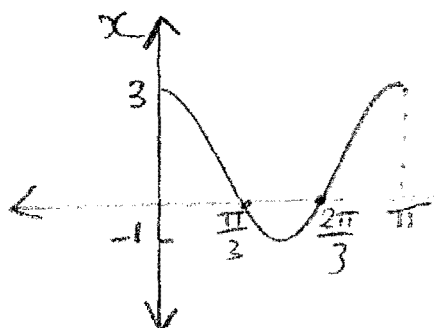
$$\therefore \ddot{x} = -4 (4 \cos^2 t - 2) \\ = -4 (4 \cos^2 t - 1 - 1) \\ = -4 (x - 1)$$

$$(ii) \quad x = 4 \cos^2 t - 1 \quad \text{but } 2 \cos 2t + 1 = 4 \cos^2 t - 1$$

$$\therefore x = 2 \cos 2t + 1$$

$$\cos 2t = -\frac{1}{2} \\ 2t = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$t = \frac{\pi}{3}, \frac{2\pi}{3}$$



>
+

$$(iii) \quad \text{Increasing most rapidly when } x = -1 \text{ (Min. value)} \\ \ddot{x} = -4(-1-1) \\ = 8 \text{ ms}^{-2} \\ t = \frac{\pi}{2}$$

$$(c) f(x) = 2x^2 \cos^{-1} 2x$$

$$\text{let } u = 2x$$

$$\frac{du}{dx} = 2$$

$$f'(x) = 2x^2 \cdot \frac{-2}{\sqrt{1-4x^2}} + (\cos^{-1} 2x) \cdot 4x$$

$$= \frac{-4x^2}{\sqrt{1-4x^2}} + 4x \cos^{-1} 2x$$

$$\text{let } y = \cos^{-1} u$$

$$\frac{dy}{du} = \frac{-1}{\sqrt{1-u^2}}$$

$$= \frac{-1}{\sqrt{1-4x^2}}$$

$$\frac{dy}{dx} = \frac{-2}{\sqrt{1-4x^2}}$$

Question - 7

$$(a) P(x) = 2x^3 - 5x^2 - 3x + 1$$

$$(i) 3(\alpha + \beta + \gamma) - 4\alpha\beta\gamma$$

$$3\left(\frac{5}{2}\right) - 4\left(-\frac{1}{2}\right)$$

$$= \frac{15}{2} + \frac{4}{2}$$

$$= \frac{19}{2}$$

$$(ii) \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

$$= \frac{-\frac{3}{2}}{-\frac{1}{2}}$$

$$= -\frac{3}{2} \times -\frac{2}{1}$$

$$= 3$$

$$\begin{aligned}
 \text{(iii)} \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\
 &= \left(\frac{5}{2}\right)^2 - 2\left(\frac{-3}{2}\right) \\
 &= \frac{37}{4}
 \end{aligned}$$

$$\text{(b)} \quad \ddot{x} = 0$$

$$\dot{x} = \int 0 dt$$

$$= C$$

$$\therefore \dot{x} = v \cos \alpha$$

$$x = \int v \cos \alpha dt$$

$$= vt \cos \alpha + C$$

$$\text{when } t=0 \quad x=0$$

$$\therefore x = vt \cos \alpha$$

$$\ddot{y} = -10$$

$$\dot{y} = \int -10 dt$$

$$= -10t + C$$

$$\text{when } t=0 \quad \dot{y} = v \sin \alpha$$

$$\therefore \dot{y} = -10t + v \sin \alpha$$

$$y = \int -10t + v \sin \alpha dt$$

$$= -5t^2 + vt \sin \alpha + C$$

$$\text{when } t=0 \quad y=0$$

$$\therefore y = -5t^2 + vt \sin \alpha$$

$$(ii) \quad \tan B = \frac{y}{x}$$

$$= \frac{V + \sin \alpha - 5t^2}{V + \cos \alpha} \quad \text{--- (1)}$$

At greatest height $y = 0$

$$10t = V \sin \alpha$$

$$t = \frac{V \sin \alpha}{10}$$

$$\therefore \text{ at Point P } x = \frac{V^2 \sin \alpha \cos \alpha}{10}$$

Returning to (1)

$$\tan B = \frac{V^2 \sin^2 \alpha}{10} = \frac{5V^2 \sin^2 \alpha}{100}$$

$$\frac{V^2 \sin \alpha \cos \alpha}{10}$$

$$= \frac{\cancel{V^2} \cancel{\sin \alpha} \cancel{V}}{20} \times \frac{10}{\cancel{V^2} \cancel{\sin \alpha} \cos \alpha}$$

$$= \frac{1}{2} \frac{\sin \alpha}{\cos \alpha}$$

$$= \frac{1}{2} \tan \alpha$$

$$(iii) \tan B = \frac{80}{120}$$

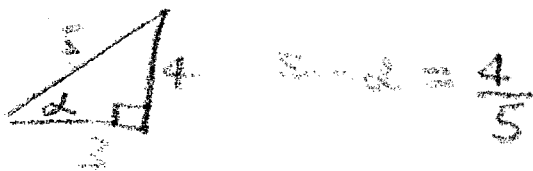
$$\tan B = \frac{2}{3}$$

$$\therefore \frac{1}{2} \tan \alpha = \frac{2}{3}$$

$$\tan \alpha = \frac{4}{3}$$

At greatest height

$$y = \frac{1}{20} v^2 \sin^2 \alpha \quad (\text{from part (ii)})$$



$$\sin \alpha = \frac{4}{5}$$

$$\therefore 80 = \frac{1}{20} \times v^2 \left(\frac{4}{5} \right)^2$$

$$80 = \frac{1}{20} \times v^2 \times \frac{16}{25}$$

$$2500 = v^2$$

$$v = 50$$

$$\therefore v = 50$$

$$\alpha = \tan^{-1} \left(\frac{4}{3} \right)$$